

Bibesh “Kumar” Timalsina

San Marcos, Texas 78666 | 737-213-2048 | timalsinabibesh747@gmail.com | Open to Relocation

[LinkedIn](#) | [GitHub](#) | [Website](#)

EDUCATION

Texas State University: College of Science and Engineering
BS, Computer Science and Data Analytics.

San Marcos, Texas
Jan 2024 – Dec 2027

Relevant Coursework:

Software Engineering, Algorithm and Analysis, Machine Learning, Artificial Intelligence, Computer Systems Security, Parallel Programming, Object Oriented Programming, Computer Architecture

EXPERIENCE

Texas State University Libraries - San Marcos, Texas

Systems Support Assistant

Dec 2025 – Present

- Collaborated with two senior software engineers to build Open API specs, controllers and services in a MVC backend for the University Libraries Travel App project used by 100+ library employees; admin feedback measured a ~30% faster processing time for travel requests.
- Built a Continuous Integration (CI) pipeline with linters, dependency checks and 14 unit tests for University Libraries Travel App project.
- Rebuilt a legacy inventory management tool as a web application secured behind TXST's SSO. The application is built as 4 Docker container services: Nginx Load Balancer, Next.js Frontend, Flask Backend and PostgreSQL database and deployed to TXST's Red Hat Enterprise Linux server via GitHub Actions CI/CD. This web application reduced departmental onboarding time by ~30% and data duplication errors by ~3%.
- Traced a data duplication error in a live Flask backend server and fixed it using WHERE NOT EXISTS subqueries.
- Built a reusable JWT cookie-based authentication system with a custom Next.js proxy and Argon2 client-side hashing. This system is used by 3 active projects.
- Built four custom reusable error pages (404, 500, 502, 504) for Texas State University Libraries using HTML and Tailwind CSS.

Texas State University Libraries, San Marcos, Texas.

STEM Research Coach

Feb 2024 – Dec 2025

- Coached 700+ undergraduates on database search strategy, research paper evaluation and locating peer-reviewed literature for stem coursework and capstone projects.
- Advised three graduate students on machine learning research projects.
- Co-authored a Research Coach FAQ with three colleagues in the university libraries CMS, creating reference for repetitive questions during advising periods.

PROJECTS

SightX: A Diabetic Retinopathy Detection System | PyTorch, FastAPI, React, Node.js, Docker, AWS | [GitHub](#) | [Blog](#)

Feb 2026 – Present

- Trained a ResNet-50 V2 classifier on 35K retinal fundus images to kappa = 0.857 on a local Apple M4 with zero cloud compute, then secured the inference with a 108-pass test-time augmentation ensemble and a cost matrix that penalizes false negatives
- Built and deployed a 3-container Docker stack (React, Node.js, FastAPI) to a Red Hat Enterprise Linux Server and Supabase as the database.
- Ran a two-week agile development cycle for model release, versioned each modal release and documented the process in my build blog.
- Currently migrating to AWS using AWS CloudFront, Lambda, SQS, RDS, S3, IAM and VPC and testing builds using Local Stack.

GitGud: A Containerized Inventory Platform | Node.js, PostgreSQL, Nginx, Docker, RHEL | [GitHub](#)

Dec 2025 – Apr 2026

- Packaged the system into 3 connected container services (Nginx, Node.js, PostgreSQL) to prevent environment drift between machines, and secured REST API routes against brute-force and token-reuse attacks with Argon2 hashing plus peppering, JWT auth, and an in-memory token blacklist.
- Wrote a GitHub Actions CI/CD pipeline that runs Jest unit tests on every Pull Request and merge to main.
- Built a Material UI dashboard with live search, status filters, and CSV export.
- Incorporated a Nginx load balancer to split traffic across replicas of a stateless backend server.

VOLUNTEERING

Association of Statistics and Analytics Student Club

IT Support and Web Developer | [Link](#) | [GitHub](#)

Aug 2026 – Present

- Primary developer and maintainer of the club's public website that has four pages covering what ASA is, membership, sign-up, and meeting details.
- The website is built with Next.js, React 19, Typescript, shadcn/ui, and Framer Motion / anime.js and deployed to Vercel and GitHub Pages with a CI/CD pipeline that separates development(dev) and production(master) branch deployments.

TECHNICAL SKILLS

Programming Languages : Python, TypeScript, SQL, C++

Frameworks, Libraries & APIs : React, Next.js, Node.js, Flask, Fast API, REST API's, WebSocket's, Open API

Databases : PostgreSQL, MongoDB, Vector DB

Cloud : AWS (CloudFront, Lambda, SQS, RDS, S3, IAM, VPC, Bedrock), Red Hat Enterprise Linux, Vercel, Supabase

Infrastructure & DevOps : Linux, GitHub Actions, NGINX, Docker, Deployment, Observability

AI/ML : Pytorch, Model Training and Serving, Transfer Learning, RAG, Vector search, Evaluations, MCP

Tools and Practices : Git, GitHub, Bitbucket, Jira, Postman, Swagger UI, Agile Development